A PROPER MORPHISM CARRIES RATIONALLY EQUIVALENT CYCLES TO RATIONALLY EQUIVALENT CYCLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Local finiteness of the images of a locally finite family. Same chapter chow.tex, label lemma-quasi-compact-locally-finite, invoked at proof line 3247. Makes the family of closed subschemes W'_j obtained as images of the W_j locally finite in Y, so that the identity to be proved can be checked one j at a time.
- Compatibility of proper pushforward with composition. Same chapter chow.tex, label lemma-compose-pushforward, invoked at proof line 3263. Rewrites p_* i_* div(f) as i'_* (p')_* div(f) for the cartesian square induced by W and its image, reducing the problem to the image.
- The principal divisor of a rational function on a proper curve over a field is zero, and pushforward along an alteration. Same chapter chow.tex, labels lemma-curve-principal-divisor (proof line 3298) and lemma-proper-pushforward-alteration (proof line 3302). The first supplies the vanishing of the multiplicity in the dimension k case through the proper integral curve W_eta over the residue field of the generic point, and the second handles the case where the image has the same dimension as W.

Source reference. Original lemma 02R5: sources/stacks-chow/pinned-original-source.tex, source lines 1924–1924. Navigation pointer for \ref{lemma-compose-pushforward} used inside the counted span (line 3264); the argument that consumes it is retained in the span.

Source reference. Original lemma 02RU: sources/stacks-chow/pinned-original-source.tex, source lines 2802–2802. Navigation pointer for \ref{lemma-curve-principal-divisor} used inside the counted span (line 3298); the argument that consumes it is retained in the span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02RT: sources/stacks-chow/pinned-original-source.tex, source lines 2645–2645. Navigation pointer for \ref{lemma-proper-pushforward-alteration} used inside the counted span (line 3302); the argument that consumes it is retained in the span.

Source reference. Original lemma 02R2: sources/stacks-chow/pinned-original-source.tex, source lines 1847–1847. Navigation pointer for \ref{lemma-quasi-compact-locally-finite} used inside the counted span (line 3247); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 3218); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X, Y be schemes locally of finite type over S. Let $p : X \to Y$ be a proper morphism. Suppose $\alpha, \beta \in Z_k(X)$ are rationally equivalent. Then $p_*\alpha$ is rationally equivalent to $p_*\beta$.*

Proof. What do we have to show? Well, suppose we are given a collection

$$i_j : W_j \longrightarrow X$$

of closed immersions, with each W_j integral of δ-dimension $k + 1$ and rational functions $f_j \in R(W_j)^*$. Moreover, assume that the collection $\{i_j(W_j)\}_{j \in J}$ is locally finite on X. Then we have to show that

$$p_* \left(\sum i_{j,*} \mathrm{div}(f_j) \right)$$

is rationally equivalent to zero on X.

Note that the sum is equal to

$$\sum p_* i_{j,*} \mathrm{div}(f_j).$$

Let $W'_j \subset Y$ be the integral closed subscheme which is the image of $p \circ i_j$. The collection $\{W'_j\}$ is locally finite in Y by Lemma 02R2. Hence it suffices to show, for a given j, that either $p_* i_{j,*} \mathrm{div}(f_j) = 0$ or that it is equal to $i'_{j,*} \mathrm{div}(g_j)$ for some $g_j \in R(W'_j)^*$.

The arguments above therefore reduce us to the case of a single integral closed subscheme $W \subset X$ of δ-dimension $k + 1$. Let $f \in R(W)^*$. Let $W' = p(W)$ as above. We get a commutative diagram of morphisms

$$\begin{array}{ccc} W & \xrightarrow{i} & X \\ {\scriptstyle p'}\downarrow & & \downarrow{\scriptstyle p} \\ W' & \xrightarrow{i'} & Y \end{array}$$

Note that $p_* i_* \mathrm{div}(f) = i'_*(p')_* \mathrm{div}(f)$ by Lemma 02R5. As explained above we have to show that $(p')_* \mathrm{div}(f)$ is the divisor of a rational function on W' or zero. There are three cases to distinguish.

The case $\dim_\delta(W') < k$. In this case automatically $(p')_* \mathrm{div}(f) = 0$ and there is nothing to prove.

The case $\dim_\delta(W') = k$. Let us show that $(p')_* \mathrm{div}(f) = 0$ in this case. Let $\eta \in W'$ be the generic point. Note that $c : W_\eta \to \mathrm{Spec}(K)$ is a proper integral curve over

$K = \kappa(\eta)$ whose function field $K(W_\eta)$ is identified with $R(W)$. Here is a diagram

$$\begin{array}{ccc} W_\eta & \longrightarrow & W \\ {\scriptstyle c}\downarrow & & \downarrow{\scriptstyle p'} \\ \mathrm{Spec}(K) & \longrightarrow & W' \end{array}$$

Let us denote $f_\eta \in K(W_\eta)^*$ the rational function corresponding to $f \in R(W)^*$. Moreover, the closed points ξ of W_η correspond $1-1$ to the closed integral subschemes $Z = Z_\xi \subset W$ of δ-dimension k with $p'(Z) = W'$. Note that the multiplicity of Z_ξ in $\mathrm{div}(f)$ is equal to $\mathrm{ord}_{\mathcal{O}_{W_\eta,\xi}}(f_\eta)$ simply because the local rings $\mathcal{O}_{W_\eta,\xi}$ and $\mathcal{O}_{W,\xi}$ are identified (as subrings of their fraction fields). Hence we see that the multiplicity of $[W']$ in $(p')_*\mathrm{div}(f)$ is equal to the multiplicity of $[\mathrm{Spec}(K)]$ in $c_*\mathrm{div}(f_\eta)$. By Lemma 02RU this is zero.

The case $\dim_\delta(W') = k+1$. In this case Lemma 02RT applies, and we see that indeed $p'_*\mathrm{div}(f) = \mathrm{div}(g)$ for some $g \in R(W')^*$ as desired. □